Airbnb Power BI Project Steps

1. Imported the dataset of Airbnb listings and reviews from csv data source.
2. Established the relationship between listings and reviews table with listings id as being the common column between two tables.
3. Created an overview dashboard report page
4. Fetched few of the important KPI's – Total listings, Total cities, Total hosts, total number of distinct property types and total number of reviews.
5. Wrote DAX logic to find the total number of listings over the years on the basis of room types- Entire place, Hotel room, Private room, Shared room using CALCULATE function.
6. Created a line chart to display the total listings on the basis of room category over the years i.e. from 2008 to 2021.
7. Created a new dashboard Ratings analysis.
8. Created a line and stacked column chart to witness the city wise listing proportions. Used DAX measures such as total listings, cumulative listings, cumulative percentage in listings to analyse the trend of the listing share amongst the cities throughout the world.
9. Created a stacked bar chart displaying the average prices of all room types in Airbnb. For the execution of this chart, I created Dax measure average price from the price column in listings table.
10. Now we created two type of charts- City wise Overview rating chart and City wise detailed rating chart overlapping with each other. To resolve this, I created two bookmarks overview and detailed which displayed alternating visual after the selection of the bookmarks
11. Created another dashboard of review analysis. Created 3 visualisations in this dashboard.
12. The first one is the review share; I used stacked column and line chart which displayed the number of reviews written by each reviewer and line showed the trend of the review proportion regarding the number of reviews in percentage. To execute it correctly, I created DAX measures and calculated columns. Dax measures included Total reviewers, Cumulative reviewers, Cumulative reviewers in percentage. Calculated Columns included Reviews per reviewer and show in review frequency chart. I learnt a lot of concepts especially DAX with this one chart such as – IF logic with OR operator, Functions such as CALCULATE, DIVIDE, ALLEXCEPT, FILTER, ALL etc.
13. The second one is the matrix kind of chart which displayed the trust factor on the basis of verification of the hosts. It showed the host bifurcation into four parameters verified, not verified, has profile and host without a profile. For the execution, I created eight Dax measures using CALCULATE function to calculate the total number of hosts with the filter condition on the basis of parameters mentioned above and after that I computed their percentages using DIVIDE function.

14. The third one is the seasonality month wise review proportion of top 5 cities using Ribbon chart. For the execution of this chart, I used DAX measures and calculated columns. I created two calculated columns- Month name and Month number and then created a DAX measure percentage of total monthly reviews by dividing total reviews with the total reviews of all the cities using CALCULATE and ALLSELECTED function in the filter section.